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Flat Weft-Knitted Upper for Sports Shoes

Abstract

Described are uppers for a sports shoe with flat weft-knitted knitwear. The flat weft-knitted knitwear forms a top portion and a bottom portion of the upper. The top portion is formed as tube weft-knitted knitwear such that it is configured to surround a part of a shank of a wearer of the sports shoe when worn. The bottom portion is configured to cover at least a part of a foot of the wearer of the sports shoe when worn.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation patent application of U.S. application Ser. No. 18/511,958, filed Nov. 16, 2023, entitled FLAT WEFT-KNITTED UPPER FOR SPORTS SHOES (“the '958 application”), which is a continuation patent application of U.S. application Ser. No. 17/668,499, filed Feb. 10, 2022, now U.S. Pat. No. 11,849,796, entitled FLAT WEFT-KNITTED UPPER FOR SPORTS SHOES (“the '499 application”), which is a continuation patent application of U.S. application Ser. No. 16/575,961, filed Sep. 19, 2019, now U.S. Pat. No. 11,272,754, entitled FLAT WEFT-KNITTED UPPER FOR SPORTS SHOES (“the '961 application”), which is a continuation patent application of U.S. application Ser. No. 14/873,605, filed Oct. 2, 2015, now U.S. Pat. No. 10,455,885, entitled FLAT WEFT-KNITTED UPPER FOR SPORTS SHOES (“the '605 application”), which is related to and claims priority benefits from German Patent Application No. DE 10 2014 220 087.3, filed on Oct. 2, 2014, entitled FLAT WEFT-KNITTED UPPER FOR SPORTS SHOES (“the '087 application”). The '958, '499, '961, '605, and '087 applications are hereby incorporated herein in their entireties by this reference.

FIELD OF THE INVENTION

[0002] The present invention relates to a flat weft-knitted upper for a shoe, in particular for sports shoes.

BACKGROUND

[0003] A number of requirements are imposed on a sports shoe, such as a running shoe, soccer shoe, basketball shoe, American Football shoe, baseball shoe or tennis shoe. This particularly includes the fact that a sports shoe is to provide very good support to the person wearing it. This is particularly important in sports with longitudinal or lateral accelerations, such as running, tennis or soccer. However, good support by the footwear is also important in sports such as climbing.

[0004] A further requirement imposed on a sports shoe is the lowest weight possible. This is particularly important when running and during fast sprints, as occur in tennis or soccer, for example.

[0005] It is currently difficult to simultaneously realize the mentioned requirements of “good support” on the one hand and “low weight” on the other hand in conventional sports shoes. Thus, a reduction of a sports shoe's weight usually results in the sports shoe providing the wearer with less support since material is dispensed with which would otherwise support the foot and ensure a firm coupling of the sports shoe to the foot.

[0006] On the other hand, the improvement of the support provided by a sports shoe to the wearer usually results in an increase in weight, for example by the application of additional fastening elements, such as buckles or hook-and-loop fasteners, or by additional reinforcements in certain

areas.

[0007] Thus, there is tension between the mentioned requirements imposed on a soccer shoe so that, at best, solutions which do meet one of the mentioned requirements while disregarding other requirements are known to date.

[0008] The present invention is therefore based on the problem of providing a sports shoe which is light on the one hand and provides very good support to a wearer on the other hand.

SUMMARY

[0009] The terms “invention,” “the invention,” “this invention” and “the present invention” used in this patent are intended to refer broadly to all of the subject matter of this patent and the patent claims below. Statements containing these terms should be understood not to limit the subject matter described herein or to limit the meaning or scope of the patent claims below. Embodiments of the invention covered by this patent are defined by the claims below, not this summary. This summary is a high-level overview of various embodiments of the invention and introduces some of the concepts that are further described in the Detailed Description section below. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this patent, any or all drawings and each claim.

[0010] According to certain embodiments of the present invention, an upper for a sports shoe comprises flat weft-knitted knitwear, wherein the flat weft-knitted knitwear forms a top portion and a bottom portion of the upper, wherein the top portion is formed as tube weft-knitted knitwear such that it is configured to surround a part of a shank of a wearer of the sports shoe when worn, and wherein the bottom portion is configured to cover at least a part of a foot of the wearer of the sports shoe when worn.

[0011] In some embodiments, the top portion and the bottom portion are connected to one another by linking and/or by weft-knitting. In further embodiments, the top portion and the bottom portion are connected to one another by ultrasonic welding. The welded seam may be covered by a band of adhesive material.

[0012] In certain embodiments, the knitwear is formed as single-surface knitwear.

[0013] According to some embodiments, the top portion and the bottom portion are adjacent to one another below an ankle of the wearer of the sports shoe when worn.

[0014] In some embodiments, knitwear is manufactured on a flat weft-knitting machine with two beds of needles. The bottom portion may be formed as two-ply knitwear.

[0015] The top portion may further comprise a weft-knitted-in pocket for a shin guard.

[0016] In some embodiments, the bottom portion comprises at least one lace bar, formed integrally with the knitwear, with at least one lace eyelet. The at least one lace bar may be formed as a layer of the knitwear. The bottom portion may comprise two lace bars, and the knitwear may be more elastic in an area between the two lace bars than in other areas.

[0017] In some embodiments, the upper further comprises a front portion not formed as knitwear. The front portion may comprise leather or artificial leather.

[0018] In certain embodiments, the top portion, at its upper edge, may comprise an elastic cuff formed integrally with the knitwear. The top portion may be adjusted to anatomical conditions of the shank of a wearer of the sports shoe and/or may be tapered from an upper edge to an ankle area. In certain embodiments, the top portion is elastic and exerts pressure on at least a part of the shank. The exerted pressure may be adjusted to the sport for which the sports shoe is used and/or may be adjusted to the respective wearer of the sports shoe. In some embodiments, the top portion comprises an elastic yarn.

[0019] According to certain embodiments, a sports shoe comprises an upper as described above and a sole connected to the upper.

[0020] According to certain embodiments of the present invention, a method of manufacturing a

shoe upper comprising flat weft-knitted knitwear comprises flat-knitting the knitwear such that the flat weft-knitted knitwear forms a top portion and a bottom portion of the upper, forming the top portion as tube weft-knitted knitwear such that it is configured to surround a part of a shank of a wearer of the sports shoe when worn, and forming the bottom portion such that it is configured to cover at least a part of a foot of the wearer of the sports shoe when worn.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] In the following detailed description, embodiments of the invention are described referring to the following figures:

[0022] FIG. **1a** is a schematic representation of textile structures, according to certain embodiments of the present invention.

[0023] FIG. **1b** is a schematic representation of a weft-knitted fabric with a filler yarn, according to certain embodiments of the present invention.

[0024] FIG. **2** are schematic representations of various interlaces of a warp-knitted fabric, according to certain embodiments of the present invention.

[0025] FIG. **3** are schematic representations of weft-knitted fabrics, according to certain embodiments of the present invention.

[0026] FIG. **4** are illustrations showing a process of stitch forming by latch needles during weft-knitting, according to certain embodiments of the present invention.

[0027] FIG. **5a** is a side view of an upper with two connected textile areas, according to certain embodiments of the present invention.

[0028] FIG. **5b** is a side view of an upper with two connected textile areas, according to certain embodiments of the present invention.

[0029] FIGS. **6a-6c** are cross-sectional views of an upper connected to a shoe sole via adhesive tape, according to certain embodiments of the present invention.

[0030] FIG. **7** are cross-sectional views of fibers for yarns used in knitwear, according to certain embodiments of the present invention.

[0031] FIG. **8** is a front view and a back view of a knitwear, according to certain embodiments of the present invention.

[0032] FIG. **9** is perspective view of an upper, according to certain embodiments of the present invention.

[0033] FIG. **10** is a side view of a sports shoe with an upper, according to certain embodiments of the present invention.

[0034] FIG. **11** is a side view of a sports shoe with an upper, according to certain embodiments of the present invention.

[0035] FIG. **12** is a top view of an upper, according to certain embodiments of the present invention.

[0036] FIG. **13a** is a top view of an upper, according to certain embodiments of the present invention.

[0037] FIG. **13b** is an inside view of the upper of FIG. **13a**.

[0038] FIG. **13c** is a side view of the upper of FIG. **13a**.

[0039] FIG. **14** is a schematic side view of an upper, according to certain embodiments of the present invention.

[0040] FIG. **15** is a schematic view of an upper, according to certain embodiments of the present invention.

[0041] FIG. **16** is a schematic view of an upper, according to certain embodiments of the present invention.

[0042] FIG. 17 is a schematic view of an upper, according to certain embodiments of the present invention.

BRIEF DESCRIPTION

[0043] The mentioned problem is solved by an upper for a sports shoe, comprising flat weft-knitted knitwear, wherein the flat weft-knitted knitwear forms a top portion and a bottom portion of the upper, wherein the top portion is formed as tube weft-knitted knitwear such that it is suitable for surrounding a part of the shank of a wearer of the sports shoe, and wherein the bottom portion is suitable for covering at least a part of a foot of the wearer of the sports shoe.

[0044] According to certain embodiments of the invention, the upper comprises flat weft-knitted knitwear. This makes the sports shoe particularly light already, since knitwear has a low weight due to hollow spaces caused by the stitches and hollow spaces in the yarns.

[0045] The flat weft-knitted knitwear of the upper according to certain embodiments of the invention forms a top portion and a bottom portion of the upper. The top portion is positioned essentially above the bottom portion when the sports shoe in which the upper is used is standing. In other words, the top portion is located closer to the edge of the foot opening than the bottom portion, with the foot opening being the opening through which a foot is inserted when the sports shoe is put on. The top portion and the bottom portion can be directly adjacent to one another or they can be spaced from one another.

[0046] According to certain embodiments of the invention, the top portion is formed as tube weft-knitted knitwear such that it is suitable for surrounding a part of the shank of a wearer of the sports shoe. Tube weft-knitted knitwear is two-ply knitwear which was manufactured on a flat weft-knitting machine with at least two beds of needles and the two plies of which are only connected on the edges. Tube weft-knitted knitwear can have a constant or a variable diameter along its length. For example, the diameter of tube weft-knitted knitwear may be tapered towards one end.

[0047] By the top portion being formed as tube weft-knitted knitwear, the top portion can tightly surround a part of a shank of a wearer of the sports shoe and thus provides additional support. Moreover, tube weft-knitted knitwear does not comprise any seams. Pressure sores or chafe marks are reduced or avoided by this.

[0048] Additionally, tube weft-knitted knitwear can be ideally adjusted to anatomical conditions. For example—in contrast to circular weft-knitted knitwear—the width, i.e. the diameter of the tube weft-knitted knitwear can be varied along its length. Due to this, it is possible to take into account the fact that the human shank is usually tapered towards the ankle. The top portion can then be formed such that it exerts essentially, i.e. palpably, constant pressure on the shank along its length. Moreover, tube weft-knitted knitwear can be asymmetrical so as to be able to even better adjust to the anatomical conditions.

[0049] The bottom portion of the upper according to the invention is suitable for covering at least a part of a foot of the wearer of the sports shoe. In certain embodiments of the invention, the bottom portion covers the foot essentially completely, i.e. from the toes to below the ankle, for example.

[0050] In summary, advantages according to certain embodiments of the invention may be achieved by using flat weft-knitted knitwear in the top and bottom portions, with the knitwear being tube weft-knitted in the bottom portion.

[0051] In some embodiments of the invention, the top portion and the bottom portion are connected to one another by linking and/or by weft-knitting (e.g. flat weft-knitting). In linking, two edges of knitwear are connected to each other according to the stitches (usually stitch by stitch). Due to this, no seam or at most a seam which only adds a little thickness is created at the line connecting the top portion and the bottom portion. Pressure sores or chafe marks are avoided or at least reduced by this. The same applies with regard to the alternative connection by weft-knitting, in case of which the top portion and the bottom portion are formed as single-surface knitwear.

[0052] Additionally or alternatively, the top portion and the bottom portion are connected to one another by sewing, gluing and/or welding.

[0053] In some embodiments of the invention, the top portion and the bottom portion are connected to one another by ultrasonic welding. Ultrasonic welding enables a simple and cost-efficient connection.

[0054] In certain embodiments of the invention, the welded seam is covered by a band of adhesive material (e.g. thermoplastic or duroplastic (thermoset) material or adhesive). The band can be arranged on the inside of the upper. In this way, the band avoids or reduces pressure sores or chafe marks. Alternatively or additionally, the band can be arranged on the outside of the upper. This can improve the optical appeal of the upper.

[0055] In certain embodiments of the invention, the knitwear is formed as single-surface knitwear. In this case, the top portion and the bottom portion are weft-knitted on a flat weft-knitting machine in one go and connected to one another in the process. This avoids the additional working step of connecting the top portion to the bottom portion.

[0056] In certain embodiments of the invention, the top portion and the bottom portion are adjacent to one another below the ankle of the wearer of the sports shoe. The top portion can be adjusted in this way in order to tightly surround the ankle of the wearer in order to protect it on the one hand and to counteract twisting of the foot on the other hand.

[0057] In certain embodiments of the invention, the knitwear is manufactured on a flat weft-knitting machine with two beds of needles. Due to this, the top portion can be weft-knitted as a tube in a simple manner.

[0058] In certain embodiments of the invention, the bottom portion is formed as two-ply knitwear. In this way, the upper can be provided with additional stability in the area of the foot.

[0059] In certain embodiments of the invention, the top portion comprises a weft-knitted-in pocket. This pocket can be filled with a shin guard, padding, an insulating layer, a warming or cooling pack and/or a sensor (particularly for communication with a mobile device), for example, and/or be used as a storage facility (e.g. for a key or money). By the pocket being weft-knitted into the top portion, it can be formed in one working step as the top portion is weft-knitted. A separate working step in which the pocket is attached can be done without.

[0060] In certain embodiments of the invention, the bottom portion comprises a weft-knitted-in pocket. This pocket can be filled with padding and/or a warming or cooling pack, for example.

[0061] In certain embodiments of the invention, the bottom portion comprises at least one lace bar, formed integrally with the knitwear, with at least one lace eyelet. Due to this, separately attaching a lace bar, for example by sewing on or gluing together, can be done without.

[0062] In certain embodiments of the invention, the at least one lace eyelet is weft-knitted into the lace bar. Thus, the lace eyelet is formed as the lace bar is weft-knitted. Subsequently forming the lace eyelet, for example by punching, can be done without and fraying of the lace eyelet can be avoided or at least reduced.

[0063] In certain embodiments of the invention, the lace bar is formed as one ply of the knitwear. In a further embodiment of the invention, a second ply of the knitwear assumes the function of a tongue. In this way, the lace bar and the tongue can be formed in one working step as the bottom portion is weft-knitted.

[0064] In certain embodiments of the invention, the bottom portion comprises two lace bars and the knitwear is more elastic in an area between the two lace bars than in other areas. In this way, the upper can adjust to different foot widths and the sports shoe can be laced up without creases being formed under the laces, since the area between the two lace bars is elastic.

[0065] In certain embodiments of the invention, the upper comprises a front portion which is not formed as knitwear. In this way, the upper can be designed depending on the requirements and it can comprise, in the front portion, a more rigid material or a material with a particular surface finish, for example.

[0066] In certain embodiments of the invention, the front portion comprises leather or artificial leather. Leather or artificial leather can be advantageous in soccer shoes in order to increase friction

("grip") with the ball.

[0067] In certain embodiments of the invention, the top portion, at its upper edge, comprises an elastic cuff formed integrally with the knitwear. The elastic cuff prevents or reduces sliding down of the top portion. By the cuff being formed integrally with the knitwear, an additional working step of connecting the cuff to the knitwear is omitted.

[0068] In certain embodiments of the invention, the cuff is weft-knitted as single Jersey. In this way, the cuff can be provided with elasticity in a simple manner. However, the cuff can also be weft-knitted as rib fabric front or purl fabric.

[0069] In certain embodiments of the invention, the top portion is adjusted to the anatomical conditions of a shank of a wearer of the sports shoe. This can take the fact into consideration that the human shank does not have a constant circumference but is usually tapered towards the ankle.

[0070] In certain embodiments of the invention, the top portion is tapered from an upper edge towards an ankle area. Thus, the top portion is ideally adjusted to the human anatomy and the top portion exerts essentially, i.e. palpably, constant pressure on the shank along its length. Pressure sores or chafe marks are avoided or reduced by this.

[0071] The invention principally allows an individual adjustment of the style and design of the sports shoe. For example, a custom-made sports shoe can be obtained using initial parameters, which have been obtained by a body scan, for example, and which represent the circumference of a leg, a calf or an ankle joint, for example. For this purpose, a standard weft-knitting pattern can be started out from, for example, and stitches can then be omitted or added depending on the initial parameters.

[0072] Moreover, the invention allows for individual designs of a sports shoe by the use of knitwear which can be formed with individual, colored yarns, graphics, logos, patterns, etc.

[0073] In certain embodiments of the invention, the top portion is elastic and exerts pressure on at least a part of the shank. The support of the sports shoe is improved by the pressure. On the other hand, a certain amount of pressure by pieces of apparel is desirable in sports, in order to provide the athlete with feedback regarding the position of their body parts ("proprioception").

[0074] In certain embodiments of the invention, the exerted pressure is adjusted to the sport for which the sports shoe is used. In this way, the foot can, for example, be provided with better support by higher pressure in case of a tennis shoe, which is subject to high accelerations due to abrupt decelerations and accelerations. In case of a running shoe, which is usually not subject to such high accelerations as a tennis shoe is, less pressure on the shank is sufficient.

[0075] In certain embodiments of the invention, the exerted pressure is adjusted to the respective wearer of the sports shoe. Depending on their personal preferences, the wearer can decide on more or less pressure on the shank by the top portion.

[0076] In certain embodiments of the invention, the top portion comprises an elastic yarn. By an elastic yarn, the top portion can be designed elastically in a simple manner, so as to exert pressure on at least a part of the shank.

[0077] The invention also relates to a sports shoe which comprises an upper according to the invention and a sole connected to the upper.

[0078] Finally, the invention relates to a method for manufacturing an upper according to the invention, comprising the following steps: Flat weft-knitting the knitwear such that the flat weft-knitted knitwear forms a top portion and a bottom portion of the upper; forming the top portion as tube weft-knitted knitwear such that it is suitable for surrounding a part of the shank of a wearer of the sports shoe; and forming the bottom portion such that it is suitable for covering at least a part of a foot of the wearer of the sports shoe.

[0079] On principle, all embodiments of the invention mentioned in this description can be combined with one another, i.e. the features of one embodiment together with the features of another embodiment constitute a further embodiment of the invention.

DETAILED DESCRIPTION

[0080] The subject matter of embodiments of the present invention is described here with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

[0081] Embodiments and variations of the present invention will be described in more detail below.

[0082] The use of knitwear allows products such as an upper (also referred to as a shoe upper) or a sole of a shoe, such as an insole, strobrel sole, midsole and/or outer sole to be equipped with areas with different characteristics providing different functions with low production effort. The properties include bendability, stretchability (expressed as Young's modulus, for example), permeability to air and water, thermoconductivity, thermal capacity, moisture absorption, static friction, abrasion resistance, hardness and thickness, for example.

[0083] Various techniques are applied in order to achieve such characteristics or functions, which will be described in the following. This includes suitable techniques in manufacturing knitwear such as knitting techniques, the selection of fibers and yarns, coating the fibers, yarns or knitwear with polymer or other materials, the use of monofilaments, the combination of monofilaments and polymer coating, the application of fuse/melt yarns, and multi-layer textile material. In general, the yarns used for the manufacture of knitwear can be equipped, e.g. coated, accordingly. In addition or alternatively, the finished knitwear can be equipped accordingly.

[0084] Another aspect of providing functions concerns the specific use of knitwear for certain areas of a product, for example of an upper or a sole, and the connection of different parts by suitable connection techniques. The mentioned aspects and techniques as well as other aspects and techniques will be explained in the following.

[0085] The described techniques can be used individually or they can be combined in any manner.

Knitwear

[0086] Knitwear used in the present invention is divided into weft-knitted fabrics and single-thread warp-knitted fabrics on the one hand and multi-thread warp-knitted fabrics on the other hand. The distinctive characteristic of knitwear is that it is formed of interlocking yarn or thread loops. These thread loops are also referred to as stitches and can be formed of one or several yarns or threads.

[0087] Yarn or thread are the terms for a structure of one or several fibers which is long in relation to its diameter. A fiber is a flexible structure which is rather thin in relation to its length. Very long fibers, of virtually unlimited length with regard to their use, are referred to as filaments.

Monofilaments are yarns consisting of one single filament, that is, one single fiber.

[0088] In weft-knitted fabrics and single-thread warp-knitted fabrics, the stitch formation requires at least one thread or yarn, with the thread running in longitudinal direction of the product, i.e. substantially at a right angle to the direction in which the product is made during the manufacturing process. In multi-thread warp-knitted fabrics, the stitch formation requires at least one warp sheet, i.e. a plurality of so-called warps. These stitch-forming threads run in longitudinal direction, i.e. substantially in the direction in which the product is made during the manufacturing process.

[0089] FIG. 1a shows the basic difference between a woven fabric **10**, weft-knitted fabrics **11** and **12** and a warp-knitted fabric **13**. A woven fabric **10** has at least two thread sheets which are usually arranged at a right angle to one another. In this regard, the threads are placed above or underneath each other and do not form stitches. Weft-knitted fabrics **11** and **12** are created by weft-knitting with one thread from the left to the right, or vice versa, by interlocking stitches. View **11** shows a front view (also referred to as the front loop fabric side) and view **12** a back view (also referred to as the back loop fabric side) of a weft-knitted fabric. The front loop and back loop product sides differ in the run of the legs **14**. On the back loop fabric side **12** the legs **14** are covered in contrast to the front loop fabric side **11**.

[0090] An alternative of a weft-knitted fabric which can be used for the present invention with a so-called filler yarn **15** is shown in FIG. **1b**. A filler yarn **15** is a length of a thread placed between two wales in longitudinal direction, which is held by transverse threads of other weave elements. By the combination of the filler yarn **15** with other weave elements the properties of the weft-knitted fabric are influenced or various pattern effects are achieved. Stretchability of the weft-knitted fabric in the direction of the wales can for example be reduced by a filler yarn **15**.

[0091] Instead of or in addition to a filler yarn, a weft can also be introduced into the knitwear during weft-knitting or warp-knitting, e.g. in order to reduce elasticity of the knitwear.

[0092] Warp-knitted fabric **13** is created by warp-knitting with many threads from top down, as shown in FIG. **1a**. In doing so, the stitches of a thread are interlocked with the stitches of the neighboring threads. Depending on the pattern according to which the stitches of the neighboring threads are interlocked, one of the seven known basic connections (also referred to as “interlaces” in multi-thread warp-knitting) pillar, tricot, 2×1 plain, satin, velvet, atlas and twill are created, for example.

[0093] By way of example, the interlaces tricot **21**, 2×1 plain **22** and atlas **23** are shown in FIG. **2**. A different interlocking results depending on how the stitches of thread **24**, which is highlighted by way of example, are interlocked in the stitches of neighboring threads. In the tricot interlace **21**, the stitch-forming thread zigzags through the knitwear in the longitudinal direction and binds between two neighboring wales. The 2×1 plain interlace **22** binds in a manner similar to that of the tricot interlace **21**, but each stitch-forming warp skips a wale. In the atlas interlace **23** each stitch-forming warp runs to a turning point in a stairs-shape and then changes direction.

[0094] Stitches arranged above each other with joint binding sites are referred to as wales. FIG. **3** shows a wale as an example of a weft-knitted fabric with reference number **31**. The term wale is also used analogously in warp-knitted fabrics. Accordingly, wales run vertically through the mesh fabric. Rows of stitches arranged next to one another, as shown by way of example for a weft-knitted fabric with reference number **32** in FIG. **3** are referred to as courses. The term course is also used analogously in warp-knitted fabrics. Accordingly, courses run through the mesh fabric in the lateral direction.

[0095] Three basic weft-knitted structures are known in weft-knitted fabrics, which can be recognized by the run of the stitches along a wale. With plain, single Jersey, only back loops can be recognized along a wale on one side of the fabric and only back loops can be recognized along the other side of the product. This structure is created on a bed of needles of a knitting machine, i.e. an arrangement of neighboring knitting needles, and also referred to as single Jersey. With rib fabric, front and back loops alternate within a course, i.e. either only front or back loops can be found along a wale, depending on the side of the product from which the wale is considered. This structure is created on two beds of needles with needles offset opposite each other. With purl fabric, front and back loops alternate in one wale. Both sides of the product look the same. This structure is manufactured by latch needles as illustrated in FIG. **4** by stitch transfer. The transfer of stitches can be avoided if double latch needles are used, which comprise both a hook and a latch at each end.

[0096] An essential advantage of knitwear over weaved textiles is the variety of structures and surfaces which can be created with it. It is possible to manufacture both very heavy and/or stiff knitwear and very soft, transparent and/or stretchable knitwear with substantially the same manufacturing technique. The parameters by which the properties of the material can be influenced substantially are the pattern of weft-knitting or warp-knitting, the used yarn, the needle size or the needle distance, and the tensile strain subject to which the yarn is placed on the needles.

[0097] An advantage of weft-knitting may be that certain yarns can be weft-knitted in at freely selectable places. In this manner, selected zones can be provided with certain properties. For example, the shoe upper for the soccer shoe according to the invention can be provided with zones made from rubberized yarn in order to achieve higher static friction and thus enable the player to

better control the ball. With certain yarns being weft-knitted in at selected places, no additional elements have to be applied.

[0098] Knitwear is manufactured on machines in the industrial context. These usually comprise a plurality of needles. In weft-knitting, latch needles **41** are usually used, which each comprise a moveable latch **42**, as illustrated in FIG. **4**. This latch **42** closes the hook **43** of the needle **41** such that a thread **44** can be pulled through a stitch **45** without the needle **41** being caught on the stitch **45**. In weft-knitting, the latch needles are usually moveable individually, so that every single needle can be controlled such that it catches a thread for stitch formation.

[0099] A differentiation is made between flat weft-knitting and circular-knitting machines. In flat weft-knitting machines, a thread feeder feeds the thread back and forth along one or several beds of needles. In a circular-knitting machine, the needles are arranged in a circular manner and the thread feeding correspondingly takes place in a circular movement along one or more round beds of needles.

[0100] Instead of a single bed of needles, it is also possible for a knitting machine to comprise two parallel beds of needles. When looked at from the side, the needles of the two beds of needles may, for example, be opposite each other at a right angle. This enables the manufacture of more elaborate structures or weaves. The use of two beds of needles allows the manufacture of a one-layered or two-layered weft-knitted fabric. A one-layered weft-knitted fabric is created when the stitches generated on the first bed of needles are enmeshed with the stitches generated on the second bed of needles. Accordingly, a two-layered weft-knitted fabric is created when the stitches generated on the first bed of needles are not or only selectively enmeshed with the stitches generated on the second bed of needles and/or if they are merely enmeshed at the end of the weft-knitted fabric. If the stitches generated on the first bed of needles are loosely enmeshed only selectively with the stitches generated on the second bed of needles by an additional yarn, this is also referred to as spacer weft-knitted fabric. The additional yarn, for example a monofilament, is thus guided back and forth between two layers, so that a distance between the two layers is created. The two layers can e.g. be connected to each other via a so-called tuck-stitch.

[0101] Generally, the following weft-knitted fabrics can thus be manufactured on a weft-knitting machine with two beds of needles: If only one bed of needles is used, a one-layered weft-knitted fabric is created. When two beds of needles are used, the stitches of both beds of needles can consistently be connected to each other so that the resulting knitwear comprises a single layer. If the stitches of both beds of needles are not connected or only connected at the edge when two beds of needles are used, two layers are created. If, when using two beds of needles, the stitches of both beds of needles are connected selectively in turns by an additional thread, a spacer weft-knitted fabric is created. The additional thread is also referred to as spacer thread and it can be fed via a separate yarn feeder.

[0102] On principle, weft-knitting machines with more than two beds of needles can also be used, which allows the design of the knitwear manufactured on them to be fashioned even more flexibly.

[0103] The techniques described herein as well as further aspects of the manufacture of knitwear can be found in “Fachwissen Bekleidung”, 6th ed. by H. Eberle et al. (published with the title “Clothing Technology” in English), in “Textil-und Modelexikon”, 6th ed. by Alfons Hofer and in “Maschenlexikon”, 11th ed. by Walter Holthaus, for example.

Three-Dimensional Knitwear

[0104] Three-dimensional (3D) knitwear can also be manufactured on weft-knitting machines and warp-knitting machines, particularly on flat weft-knitting machines. This is knitwear which comprises a spatial structure although it is weft-knitted or warp-knitted in a single process. A three-dimensional weft-knitting or warp-knitting technique allows for spatial knitwear to be manufactured without seams, cut or manufacture in one piece and in a single process.

[0105] Three-dimensional knitwear may, for example, be manufactured by varying the number of stitches in the direction of the wales by the formation of partial courses. The corresponding

mechanical process is referred to as “needle parking”. Depending on the requirement, this can be combined with structural variations and/or variations of the number of stitches in the direction of the course. When partial courses are formed, stitch formation temporarily occurs only along a partial width of the weft-knitted fabric or warp-knitted fabric. The needles which are not involved in the stitch formation keep the half stitches (“needle parking”) until weft-knitting occurs again at this position. In this way, it is possible to achieve bulges, for example.

[0106] By three-dimensional weft-knitting or warp-knitting an upper can be adjusted to the cobbler's last or the foot and a sole can be treaded, for example. The tongue of a shoe can e.g. be weft-knitted into the right shape. Contours, structures, knobs, curvatures, notches, openings, fasteners, loops and pockets can be integrated into the knitwear in a single process.

[0107] Three-dimensional knitwear can be used for the present invention in an advantageous manner.

[0108] For example, a three-dimensional shoe can be weft-knitted using a single-Jersey material which extends from the topmost portion of the upper to the toes and/or another area of a bottom portion of the shoe. The topmost portion of the upper can be weft-knitted using elastic yarns with a predetermined elasticity module, whereas the bottom portion of the upper can include several yarns of different types. For example, the bottom portion of a shoe upper can include several yarns, such as yarns made from polyester, monofilament, elastic yarns and/or combinations thereof.

Functional Knitwear

[0109] Knitwear and particularly weft-knitted fabric can be provided with a range of functional properties and used in the present invention in an advantageous manner.

[0110] It is possible by a weft-knitting technique to manufacture knitwear which has different functional areas and simultaneously maintains its contours. The structures of knitwear may be adjusted to functional requirements in certain areas, by the stitch pattern, the yarn, the needle size, the needle distance or the tensile strain subject to which the yarn is placed on the needles being selected accordingly.

[0111] It is possible, for example, to include structures with large stitches or openings within the knitwear in areas in which airing is desired. In contrast, in areas in which support and stability are desired, fine-meshed stitch patterns, stiffer yarns or even multi-layered weft-knitting structures can be used, which will be described in the following. In the same manner, the thickness of the knitwear is variable.

[0112] Knitwear having more than one layer provides numerous possible constructions for the knitwear, which provide many advantages. Knitwear with more than one layer, e.g. two, can be weft-knitted or warp-knitted on a weft-knitting machine or a warp-knitting machine with several beds of needles, e.g. two, in a single stage, as described in the section “knitwear” above. Alternatively, the several layers, e.g. two, can be weft-knitted or warp-knitted in separate stages and then placed above each other and connected to each other if applicable, e.g. by sewing, gluing, welding or linking.

[0113] Several layers fundamentally increase solidness and stability of the knitwear. In this regard, the resulting solidness depends on the extent to which and the techniques by which the layers are connected to each other. The same yarn or different yarns can be used for the individual layers. For example, it is possible in a weft-knitted fabric for one layer to be weft-knitted from multi-fiber yarn and one layer to be weft-knitted from monofilament, whose stitches are enmeshed. In particular stretchability of the weft-knitted layer is reduced due to this combination of different yarns. It is an advantageous alternative of this construction to arrange a layer made from monofilament between two layers made from multi-fiber yarn in order to reduce stretchability and increase solidness of the knitwear. This results in a pleasant surface made from multi-fiber yarn on both sides of the knitwear.

[0114] An alternative of two-layered knitwear is referred to as spacer weft-knitted fabric or spacer warp-knitted fabric, as explained in the section “knitwear”. In this regard, a spacer yarn is weft-

knitted or warp-knitted more or less loosely between two weft-knitted or warp-knitted layers, interconnecting the two layers and simultaneously serving as a filler. The spacer yarn can comprise the same material as the layers themselves, e.g. polyester or another material. The spacer yarn can also be a monofilament which provides the spacer weft-knitted fabric or spacer warp-knitted fabric with stability.

[0115] Such spacer weft-knitted fabrics or spacer warp-knitted fabrics, which are also referred to as three-dimensional weft-knitted fabrics or warp-knitted fabrics, but have to be differentiated from the formative 3D weft-knitted fabrics or 3D warp-knitted fabrics mentioned in the section “three-dimensional knitwear” above, can be used wherever additional cushioning or protection is desired, e.g. at the upper or the tongue of an upper or in certain areas of a sole. Three-dimensional structures can also serve to create spaces between neighboring textile layers or also between a textile layer and the foot and thus ensure airing. Moreover, the layers of a spacer weft-knitted fabric or a spacer warp-knitted fabric can comprise different yarns depending on the position of the spacer weft-knitted fabric on the foot.

[0116] The thickness of a spacer weft-knitted fabric or a spacer warp-knitted fabric can be set in different areas depending on the function or the wearer. Various degrees of cushioning can be achieved with areas of various thicknesses, for example. Thin areas can increase bendability, for example, thus fulfilling the function of joints or flex lines.

[0117] Multi-layer constructions also create possibilities of color design, by different colors being used for the different layers. In this way, knitwear can be provided with two different colors for the front and the back, for example. An upper made from such knitwear can then comprise a different color on the outside than on the inside.

[0118] An alternative of multi-layered constructions are pockets or tunnels, in which two textile layers or knitwear weft-knitted or warp-knitted on two beds of needles are connected to each other only in certain areas so that a hollow space is created. Alternatively, items of knitwear weft-knitted or warp-knitted in two separate processes are connected to each other such that a void is created, e.g. by sewing, gluing, welding or linking. It is then possible to introduce a cushioning material such as a foam material, eTPU (expanded thermoplastic urethane), ePP (expanded polypropylene), expanded EVA (ethylene vinyl acetate) or particle foam, an air or gel cushion for example, through an opening, e.g. at the tongue, the upper, the heel, the sole or in other areas. Alternatively or additionally, the pocket can also be filled with a filler thread or a spacer knitwear. It is furthermore possible for threads to be pulled through tunnels, for example as reinforcement in case of tension loads in certain areas of an upper. Moreover, it is also possible for the laces to be guided through such tunnels. Moreover, loose threads can be placed into tunnels or pockets for padding, for example in the area of the ankle. However, it is also possible for stiffer reinforcing elements, such as caps, flaps or bones to be inserted into tunnels or pockets. These can be manufactured from plastic such as polyethylene, TPU, polyethylene or polypropylene, for example.

[0119] A further possibility for a functional design of knitwear is the use of certain variations of the basic weaves. In weft-knitting, it is possible for bulges, ribs or waves to be weft-knitted in certain areas, for example, in order to achieve reinforcement in these places. A wave may, for example, be created by stitch accumulation on a layer of knitwear. This means that more stitches are weft-knitted or warp-knitted on one layer than on another layer. Alternatively, different stitches are weft-knitted on the one layer than on the other layer, e.g. with these being weft-knitted tighter, wider or using a different yarn. Thickening is caused in both alternatives.

[0120] Ribs, waves or similar patterns may, for example, also be used at the bottom of a weft-knitted outer sole of a shoe in order to provide a tread and provide the shoe with better non-slip properties. In order to obtain a rather thick weft-knitted fabric, for example, it is possible to use the weft-knitting techniques “tuck” or “half cardigan”, which are described in “Fachwissen Bekleidung”, 6th ed. by H. Eberle et al., for example.

[0121] Waves can be weft-knitted or warp-knitted such that a connection is created between two

layers of a two-layered knitwear or such that no connection is created between the two layers. A wave can also be weft-knitted as a right-left wave on both sides with or without a connection of the two layers. A structure in the knitwear can be achieved by an uneven ration of stitches on the front or the back of the knitwear.

[0122] Ribs, waves or similar patterns, for example, may be included in the knitwear of the soccer shoe according to the invention in order to increase friction with a soccer ball, for example, and/or in order to generally allow for a soccer player to have better control of the ball.

[0123] A further possibility of functionally designing knitwear within the framework of the present invention is providing openings in the knitwear already during weft-knitting or warp-knitting. In this manner, airing of the soccer shoe according to the invention may be provided in specific places in a simple manner.

[0124] Yet another possibility of functionally designing knitwear within the framework of the present invention is forming laces integrally with the knitwear of the shoe upper according to the invention. In these embodiments, the laces are warp-knitted or weft-knitted integrally with the knitwear already when the knitwear of the shoe upper according to the invention is weft-knitted or warp-knitted. In this regard, a first end of a lace is connected to the knitwear, while a second end is free.

[0125] In certain embodiments, the first end is connected to the knitwear of the upper in the area of the transition from the tongue to the area of the forefoot of the upper. In further embodiments, a first end of a first lace is connected to the knitwear of the upper at the medial side of the tongue and a first end of a second lace is connected to the knitwear of the upper at the lateral side of the tongue. The respective second ends of the two laces can then be pulled through lace eyelets for tying the shoe.

[0126] A possibility of speeding up the integral weft-knitting or warp-knitting of laces is having all yarns used for weft-knitting or warp-knitting knitwear end in the area of the transition from the tongue to the area of the forefoot of the upper. The yarns may end in the medial side of the upper on the medial side of the tongue and form the lace connected on the medial side of the tongue. The yarns may end in the lateral side of the upper on the lateral side of the tongue and form the lace connected to the lateral side of the tongue. The yarns may then be cut off at a length which is sufficiently long for forming laces. The yarns can be twisted or intertwined, for example. The respective second end of the laces may be provided with a lace clip. Alternatively, the second ends are fused or provided with a coating.

[0127] The knitwear is particularly stretchable in the direction of the stitches (longitudinal direction) due to its construction. This stretching can be reduced e.g. by subsequent polymer coating of the knitwear. The stretching can also be reduced during manufacture of the knitwear itself, however. One possibility is reducing the mesh openings, that is, using a smaller needle size. Smaller stitches generally result in less stretching of the knitwear. Moreover, the stretching of the knitwear can be reduced by weft-knitted reinforcements, e.g. three-dimensional structures. Such structures may be arranged on the inside or the outside of the knitwear of the shoe upper according to the invention. Furthermore, non-stretchable yarn, e.g. made from nylon, can be laid in a tunnel along the knitwear in order to limit stretching to the length of the non-stretchable yarn.

[0128] Colored areas with several colors can be created by using a different thread and/or by additional layers. In transitional areas, smaller mesh openings (smaller needle sizes) are used in order to achieve a fluent passage of colors.

[0129] Further effects can be achieved by weft-knitted insets (inlaid works) or Jacquard knitting. Inlaid works are areas which only provide a certain yarn, e.g. in a certain color. Neighboring areas which can comprise a different yarn, for example in a different color, are then connected to each other by a so-called tuck-stitch.

[0130] During Jacquard knitting, two beds of needles are used and two different yarns run through all areas, for example. However, in certain areas only one yarn appears on the visible side of the

knitwear and the respective other yarn runs invisibly on the other side of the knitwear.

[0131] A product manufactured from knitwear can be manufactured in one piece on a weft-knitting machine or a warp-knitting machine. Functional areas can then already be manufactured during weft-knitting or warp-knitting by corresponding techniques as described here.

[0132] Alternatively, the product can be combined from several parts of knitwear and it can also comprise parts which are not manufactured from knitwear. In this regard, the parts of knitwear can each be designed separately with different functions, for example regarding thickness, insulation, transport of moisture, stability, protection, abrasion-resistance, durability, cooling, stretchability, rigidity, compression, etc.

[0133] The shoe upper of the soccer shoe according to the invention and/or its sole may, for example, be generally manufactured from knitwear as a whole or it may be put together from different parts of knitted goods. A whole upper or parts of that may, for example, be separated, e.g. punched, from a larger piece of knitwear. The larger piece of knitwear may, for example, be a circular weft-knitted fabric or a circular warp-knitted fabric or a flat weft-knitted fabric or a flat warp-knitted fabric.

[0134] In a further example, a shoe comprises a flat weft-knitted strobel sole, an insole and/or an outsole. The outsole can be connected to the upper by sewing, for example. Other connecting techniques may be utilized.

[0135] For example, a tongue can be manufactured as a continuous piece and connected with the upper subsequently, or it can be manufactured in one piece with the upper. With regard to their functional designs, ridges on the inside can e.g. improve flexibility of the tongue and ensure that a distance is created between the tongue and the foot, which provides additional airing. Laces may be guided through one or several weft-knitted tunnels of the tongue. The tongue can also be reinforced with polymer in order to achieve stabilization of the tongue and e.g. prevent a very thin tongue from convolving. Moreover, the tongue can then also be fitted to the shape of the cobbler's last or the foot.

[0136] Applications such as polyurethane (PU) prints, thermoplastic polyurethane (TPU) ribbons, textile reinforcements, leather, rubber, etc., may be subsequently applied to the knitwear of the soccer shoe according to the invention. Thus, it is possible, for example, to apply a plastic heel or toe cap as reinforcement or logos and eyelets for laces on the shoe upper, for example by sewing, gluing or welding, as described below.

[0137] Sewing, gluing or welding, for example, constitute suitable connection techniques for connecting individual parts of knitwear with other textiles or with parts of knitwear. Linking is another possibility for connecting two parts of knitwear. Therein, two edges of knitwear are connected to each other according to the stitches (usually stitch by stitch).

[0138] A possibility for welding textiles, particularly ones made from plastic yarns or threads, is ultrasonic welding. Therein, mechanical oscillations in the ultrasonic frequency range are transferred to a tool referred to as a sonotrode. The oscillations are transferred to the textiles to be connected by the sonotrode under pressure. Due to the resulting friction, the textiles are heated up, softened and ultimately connected in the area of the place of contact with the sonotrode. Ultrasonic welding allows rapidly and cost-effectively connecting particularly textiles with plastic yarns or threads. It is possible for a ribbon to be attached, for example glued, to the weld seam, which additionally reinforces the weld seam and is optically more appealing. Moreover, wear comfort is increased since skin irritations-especially at the transition to the tongue-are avoided.

[0139] Connecting various textile areas, such as parts of knitwear, may occur at quite different locations. For example, the seams for connecting various textile areas of the shoe upper of the soccer shoe according to the invention can be arranged at various positions, as shown in FIGS. 5a and 5b. An upper 51 is shown in FIG. 5a which comprises two textile areas 52 and 53. They are sewn to each other. The seam 54 which connects the two textile areas 52 and 53 runs diagonally from an instep area of the upper to an area of the sole in the transition from the midfoot to the

heel. In FIG. **5b** the seam **55** also runs diagonally, but it is arranged more to the front in the direction of the toes. Other arrangements of seams and connecting places in general may be utilized. The seams shown in FIGS. **5a** and **5b** can each be a thread seam, a glued seam, a welded seam or a linking seam. The two seams **54** and **55** can each be attached only on one side of the upper **51** or on both sides of the upper.

[0140] The use of adhesive tape constitutes a further possibility for connecting textile areas. This can also be used in addition to an existing connection, e.g. over a sewn seam or a welded seam. An adhesive tape can fulfill further functions in addition to the function of connecting, such as e.g. protection against dirt or water. An adhesive tape can comprise properties which change over its length.

[0141] Certain embodiments of an upper **51** connected to a shoe sole **61** by adhesive tape

[0142] is shown in FIGS. **6a**, **6b** and **6c**. Each of FIGS. **6a**, **6b** and **6c** shows a cross-section through a shoe with different positions of the foot and the deformations of the shoe caused by that. For example, tensile forces work on the right side of the shoe in FIG. **6a**, whereas compression forces work on the left side.

[0143] The shoe sole **61** can be an outer sole or a midsole. The upper **51** and the shoe sole **61** are connected to each other by a surrounding adhesive tape **62**. The adhesive tape **62** can be of varying flexibility along its length. For example, the adhesive tape **62** might be particularly rigid and not very flexible in the shoe's heel area in order to provide the shoe with the necessary stability in the heel area. This may be achieved by varying the width and/or the thickness of the adhesive tape **62**, for example. The adhesive tape **62** can generally be constructed such that it is able to receive certain forces in certain areas along the tape. In this way, the adhesive tape **62** does not only connect the upper to the sole but simultaneously fulfils the function of structural reinforcement.

Fibers

[0144] The yarns or threads used for the knitwear of the present invention usually comprise fibers. As was explained above, a flexible structure which is rather thin in relation to its length is referred to as a fiber. Very long fibers, of virtually unlimited length with regard to their use, are referred to as filaments. Fibers are spun or twisted into threads or yarns. Fibers can also be long, however, and twirled into a yarn. Fibers can consist of natural or synthetic materials.

[0145] Natural fibers are environmentally friendly, since they are compostable. Natural fibers include cotton, wool, alpaca, hemp, coconut fibers or silk, for example. Among the synthetic fibers are polymer-based fibers such as Nylon™, polyester, elastane or spandex or Kevlar®, which can be produced as classic fibers or as high-performance fibers or technical fibers.

[0146] In certain embodiments, a soccer shoe according to the invention may be assembled from various parts, with a weft-knitted or a warp-knitted part comprising natural yarn made from natural fibers and a removable part, e.g. the insole, comprising plastic, for example. In this manner, both parts can be disposed of separately. In this example, the weft-knitted part could be directed to compostable waste, whereas the insole could be directed to recycling of reusable materials, for example.

[0147] The mechanical and physical properties of a fiber and the yarn manufactured therefrom are also determined by the fiber's cross-section, as illustrated in FIG. **7**. These different cross-sections, their properties and examples of materials having such cross-sections will be explained in the following.

[0148] A fiber having the circular cross-section **710** can either be solid or hollow. A solid fiber is the most frequent type, it allows easy bending and is soft to the touch. A fiber as a hollow circle with the same weight/length ratio as the solid fiber has a larger cross-section and is more resistant to bending. Examples of fibers with a circular cross-section are Nylon™, polyester and Lyocell.

[0149] A fiber having the bone-shaped cross-section **730** has the property of wicking moisture. Examples of materials for such fibers are acrylic and spandex. The concave areas in the middle of the fiber support moisture being passed on in the longitudinal direction, with moisture being

rapidly wicked from a certain place and distributed.

[0150] The following further cross-sections are illustrated in FIG. 7: [0151] Polygonal cross-section **711** with flowers; example: flax; [0152] Oval to round cross-section **712** with overlapping sections; example: wool; [0153] Flat, oval cross-section with expansion and convolution **713**; example: cotton; [0154] circular, serrated cross-section **714** with partial striations; rayon; [0155] lima bean cross-section **720**; smooth surface; [0156] Serrated lima bean cross-section **721**, example: Avril™ rayon; [0157] triangular cross-section **722** with rounded edges; example: silk; [0158] trilobal star cross-section **723**; like triangular fiber with shinier appearance; [0159] Clubbed cross-section **724** with partial striations; sparkling appearance; example: acetate; [0160] flat and broad cross-section **731**; example: acetate in another design; [0161] star-shaped or concertina cross section **732**; [0162] cross-section **733** in the shape of a collapsed tube with a hollow center; and [0163] Square cross-section with voids **734**; example: AnsoIV™ nylon.

[0164] Individual fibers with their properties which are relevant for the manufacture of knitwear for the present invention will be described in the following: [0165] Aramid fibers: good resistance to abrasion and organic solvents; non-conductive; temperature-resistant up to 500° C.; low flammability. [0166] Para-aramid fibers: known under trade names Kevlar®, Techova™ and Twaron™. outstanding strength-to-weight properties; high Young's modulus and high tensile strength (higher than with meta-aramides); low stretching and low elongation at break (approx. 3.5%). [0167] Meta aramides: known under trade names Numex™, Teijinconex™, New Star™, X-Fiber™. [0168] Dyneema fibers: highest impact strength of any known thermoplastics; highly resistant to corrosive chemicals, with exception of oxidizing acids; extremely low moisture absorption; very low coefficient of friction, which is significantly lower than that of nylon™ and acetate and comparable to Teflon; self-lubricating; highly resistant to abrasion (15 times more resistant to abrasion than carbon steel); nontoxic. [0169] Carbon fiber: an extremely thin fiber about 0.005-0.010 mm in diameter, composed substantially of carbon atoms; highly stable with regard to size; one yarn is formed from several thousand carbon fibers; high tensile strength; low weight; low thermal expansion; very strong when stretched or bent; thermal conductivity and electric conductivity. [0170] Glass fiber: high ratio of surface area to weight; by trapping air within them, blocks of glass fibers provide good thermal insulation; thermal conductivity of 0.05 W/(m×K); the thinnest fibers are the strongest because the thinner fibers are more ductile; the properties of the glass fibers are the same along the fiber and across its cross-section, since glass has an amorphous structure; correlation between bending diameter of the fiber and the fiber diameter; thermal, electrical and sound insulation; higher stretching before it breaks than carbon fibers.

Yarns

[0171] A plurality of different yarns can be used for the manufacture of knitwear which is used in the present invention. As was already defined, a structure of one or several fibers which is long in relation to its diameter is referred to as a yarn.

[0172] Functional yarns are capable of transporting moisture and thus of absorbing sweat and moisture. They can be electrically conducting, self-cleaning, thermally regulating and insulating, flame resistant, and UV-absorbing, and can enable infrared radiation. They can be suitable for sensorics. Antibacterial yarns, such as silver yarns, for example, prevent odor formation.

[0173] Stainless steel yarn contains fibers made of a blend of nylon or polyester and steel. Its properties include high abrasion resistance, high cut resistance, high thermal abrasion, high thermal and electrical conductivity, higher tensile strength and high weight.

[0174] In textiles made from knitwear, electrically conducting yarns can be used for the integration of electronic devices. These yarns may, for example, forward impulses from sensors to devices for processing the impulses, or the yarns can function as sensors themselves, and measure electric streams on the skin or physiological magnetic fields, for example. Examples for the use of textile-based electrodes can be found in European patent application EP 1 916 323.

[0175] Fuse yarns can be a mixture of a thermoplastic yarn and a non-thermoplastic yarn. There are essentially three types of fuse/melt yarns: a thermoplastic yarn surrounded by a non-thermoplastic yarn; a non-thermoplastic yarn surrounded by thermoplastic yarn; and pure fuse/melt yarn of a thermoplastic material. After being heated to the melting temperature, the thermoplastic yarn fuses with the non-thermoplastic yarn (e.g. polyester or nylon™), stiffening the knitwear. The melting temperature of the thermoplastic yarn is determined accordingly and it is usually lower than that of the non-thermoplastic yarn in case of a mixed yarn.

[0176] A shrinking yarn is a dual-component yarn. The outer component is a shrinking material, which shrinks when a defined temperature is exceeded. The inner component is a non-shrinking yarn, such as polyester or nylon. Shrinking increases the stiffness of the textile material.

[0177] A further yarn for use in knitwear are luminescent or reflecting yarns and so-called “intelligent” yarns. Examples of intelligent yarns are yarns which react to humidity, heat or cold and alter their properties accordingly, e.g. contracting and thus making the stitches smaller or changing their volume and thus increasing permeability to air. Yarns made from piezo fibers or yarn coated with a piezo-electrical substance are able to convert kinetic energy or changes in pressure into electricity, which can provide energy to sensors, transmitters or accumulators, for example.

[0178] Yarns can furthermore generally be reworked, e.g. coated, in order to maintain certain properties, such as stretching, water resistance/water repellence, color or humidity resistance.

Polymer Coating

[0179] Due to its structure, weft-knitted or warp-knitted knitwear is considerably more flexible and stretchable than weaved textile materials. For certain applications and requirements, e.g. in certain areas of a shoe upper according to the present invention, it is therefore necessary to reduce flexibility and stretchability in order to achieve sufficient stability.

[0180] For that purpose, a polymer layer can be applied to one side or both sides of knitwear (weft-knit or warp-knit goods), but generally also to other textile materials. Such a polymer layer causes a reinforcement and/or stiffening of the knitwear. In a shoe upper in accordance with the present invention, it may, for example, serve the purpose of supporting and/or stiffening and/or reducing elasticity in the toe area, in the heel area, along the lace eyelets, on lateral and/or medial surfaces or in other areas. Furthermore, elasticity of the knitwear and particularly stretchability are reduced. Moreover, the polymer layer protects the knitwear against abrasion. Furthermore, it is possible to give the knitwear a three-dimensional shape by the polymer coating by compression-molding. The polymer coating may be thermoplastic urethane (TPU), for example.

[0181] In the first step of polymer coating, the polymer material is applied to one side of the knitwear. It can also be applied on both sides, however. The material can be applied by spraying on, coating with a doctor knife, laying on, printing on, sintering, ironing on or spreading. If it is polymer material in the form of a film, the latter is placed on the knitwear and connected with the knitwear by heat and pressure, for example. The most important method of applying is spraying on. This can be carried out by a tool similar to a hot glue gun. Spraying on enables the polymer material to be applied evenly in thin layers. Moreover, spraying on is a fast method. Effect pigments such as color pigments, for example, may be mixed into the polymer coating.

[0182] The polymer is applied in at least one layer with a thickness of approximately 0.2-1 mm. One or several layers can be applied, with it being possible for the layers to be of different thicknesses and/or colors. For example, a shoe can comprise a polymer coating with a thickness of 0.01 to 5 mm. Furthermore, with some shoes, the thickness of the polymer coating can be in the range of 0.05 to 2 mm. Between neighboring areas of a shoe with polymer coating of various thicknesses there can be continuous transitions from areas with a thin polymer coating to areas with a thick polymer coating. In the same manner, different polymer materials can be used in different areas, as will be described in the following.

[0183] During application, polymer material attaches itself to the points of contact or points of

intersection, respectively, of the yarns of the knitwear, on the one hand, and to the gaps between the yarns, on the other hand, forming a closed polymer surface on the knitwear after the processing steps described in the following. However, in case of larger mesh openings or holes in the textile structure, this closed polymer surface can also be intermittent, e.g. so as to enable airing. This also depends on the thickness of the applied material: The more thinly the polymer material is applied, the easier it is for the closed polymer surface to be intermittent. Moreover, the polymer material can also penetrate the yarn and soak it and thus contributes to its stiffening.

[0184] After application of the polymer material, the knitwear is pressed in a press under heat and pressure. The polymer material liquefies in this step and fuses with the yarn of the textile material.

[0185] In a further optional step, the knitwear can be pressed into a three-dimensional shape in a machine for compression-molding. For example the area of the heel or the area of the toes of a shoe upper can be shaped three-dimensionally over a cobbler's last. Alternatively, the knitwear can also be directly fitted to a foot.

[0186] After pressing and molding, the reaction time until complete stiffening can be one to two days, depending on the used polymer material.

[0187] The following polymer materials can be used: polyester; polyester-urethane pre-polymer; acrylate; acetate; reactive polyolefins; co-polyester; polyamide; co-polyamide; reactive systems (mainly polyurethane systems reactive with H.sub.2O or O.sub.2); polyurethanes; thermoplastic polyurethanes; and polymeric dispersions.

[0188] The described polymer coating can be used sensibly wherever support functions, stiffening, increased abrasion resistance, elimination of stretchability, increase of comfort, increase of friction and/or fitting to prescribed three-dimensional geometries are desired. In certain embodiments, the shoe upper in accordance with the present invention may be fitted to the individual shape of the foot of the person wearing it, by polymer material being applied to the shoe upper and then adapting to the shape of the foot under heat.

[0189] In addition or alternatively to a reinforcing polymer coating, a knitwear can also be equipped with a water-repellant coating in order to avoid or at least reduce the ingress of humidity into an upper, for example. In this regard, the water-repellant coating can be applied to the entire upper or only to a part thereof, e.g. in the area of the toes. Water-repellant coatings can be based on hydrophobic materials such as polytetrafluoroethylene (PTFE), wax or paraffin. A commercially available coating is Scotchgard™ by 3M.

Monofilaments for Reinforcement

[0190] As was already defined, a monofilament is a yarn consisting of one single filament, that is, one single fiber. Therefore, stretchability of monofilaments is considerably lower than that of yarns which are manufactured from many fibers. This also reduces the stretchability of knitwear which are manufactured from monofilaments or comprise monofilaments and which are used in the present invention. Monofilaments are typically made from polyamide. However, other materials, such as polyester or a thermoplastic material, may also be used.

[0191] So whereas knitwear made from a monofilament is considerably more rigid and less stretchable, this knitwear, however, does not have the desired surface properties such as e.g. smoothness, colors, transport of moisture, outer appearance and variety of textile structures as usual knitwear has. This disadvantage is overcome by the knitwear described in the following.

[0192] FIG. 8 depicts a weft-knitted fabric having a weft-knitted layer made from a first yarn, such as a multi-fiber yarn, for example, and a weft-knitted layer made from monofilament. The layer of monofilament is weft-knitted into the layer of the first yarn. The resulting two-layered knitwear is considerably more solid and less stretchable than the layer made from yarn alone.

[0193] FIG. 8 particularly depicts a front view **81** and a back view **82** of a two-layered knitwear **80**. Both views show a first weft-knitted layer **83** made from a first yarn and a second weft-knitted layer **84** made from monofilament. The first textile layer **83** made from a first yarn is connected to the second layer **84** by stitches **85**. Thus, the greater solidness and smaller stretchability of the

second textile layer **84** made from the monofilament is transferred to the first textile layer **83** made from the first yarn.

[0194] A monofilament can also be begun to be melted slightly in order to connect with the layer of the first yarn and limit stretching even more. The monofilament then fuses with the first yarn at the points of contact and fixates the first yarn with respect to the layer made from monofilament.

Combination of Monofilaments and Polymer Coating

[0195] The weft-knitted fabric having two layers described in the preceding section can additionally be reinforced by a polymer coating as was already described in the section “polymer coating”. The polymer material is applied to the weft-knitted layer made from monofilament. In doing so, it does not connect to the material (e.g. polyamide material) of the monofilament, since the monofilament has a very smooth and round surface, but substantially penetrates the underlying first layer of a first yarn (e.g. polyester yarn). During subsequent pressing, the polymer material therefore fuses with the yarn of the first layer and reinforces the first layer. In doing so, the polymer material has a lower melting point than the first yarn of the first layer and the monofilament of the second layer. The temperature during pressing is selected such that only the polymer material melts but not the monofilament or the first yarn.

Fuse Yarn

[0196] For reinforcement and for the reduction of stretching, the yarn of the knitwear which is used according to the invention can additionally or alternatively also be a melted yarn which fixes the knitwear after pressing. There are essentially three types of fuse/melt yarns: a thermoplastic yarn surrounded by a non-thermoplastic yarn; a non-thermoplastic yarn surrounded by thermoplastic yarn; and pure fuse/melt yarn of a thermoplastic material. In order to improve the bond between thermoplastic yarn and the non-thermoplastic yarn, it is possible for the surface of the non-thermoplastic yarn to be texturized.

[0197] Pressing may take place at a temperature ranging from 110 to 150° C., and may further take place at 130° C. The thermoplastic yarn melts at least partially in the process and fuses with the non-thermoplastic yarn. After pressing, the knitwear is cooled, so that the bond is hardened and fixed. The fuse yarn may be arranged in the entire knitwear or only in selective areas.

[0198] In certain embodiments, the fuse yarn is weft-knitted or warp-knitted into the knitwear. In case of several layers, the melted yarn can be weft-knitted into one, several or all layers of the knitwear.

[0199] In further embodiments, the melt yarn can be arranged between two layers of knitwear. In doing so, the melt yarn can simply be placed between the layers. Arrangement between the layers has an advantage that the mold is not stained during pressing and molding, since there is no direct contact between the melt yarn and the mold.

Thermoplastic Textile for Reinforcement

[0200] A further possibility for reinforcing knitwear which is used for the present invention is the use of a thermoplastic textile. This is a thermoplastic woven fabric or thermoplastic knitwear. A thermoplastic textile melts at least partially subject to heat and stiffens as it cools down. A thermoplastic textile may, for example, be applied to the surface of the knitwear by applying pressure and heat. When it cools down, the thermoplastic textile stiffens and specifically reinforces the shoe upper in the area in which it was placed, for example.

[0201] The thermoplastic textile can specifically be manufactured for the reinforcement in its shape, thickness and structure. Additionally, its properties can be varied in certain areas. The stitch structure, the knitting stitch and/or the yarn used can be varied such that different properties are achieved in different areas.

[0202] In certain embodiments, a thermoplastic textile is a weft-knitted fabric or warp-knitted fabric made from thermoplastic yarn. Additionally, the thermoplastic textile can also comprise a non-thermoplastic yarn. The thermoplastic textile may be applied to the shoe upper of the soccer shoe according to the invention, for example, by pressure and heat.

[0203] A woven fabric whose wefts and/or warps are thermoplastic are further embodiments of a thermoplastic textile. Different yarns can be used in the weft direction and the warp direction of the thermoplastic woven fabric, so as to achieve different properties, such as stretchability, in the weft direction and the warp direction.

[0204] A spacer weft-knitted fabric or spacer warp-knitted fabric made from thermoplastic material are further embodiments of a thermoplastic textile. In this regard, only one layer may be thermoplastic, for example, e.g. so as to be attached to the shoe upper of the soccer shoe according to the invention. Alternatively, both layers are thermoplastic, e.g. in order to connect the sole to the upper.

[0205] A thermoplastic weft-knitted fabric or warp-knitted fabric can be manufactured using the manufacturing techniques for knitwear described in the section “knitwear”.

[0206] A thermoplastic textile can be connected with the surface to be reinforced only partially subject to pressure and heat so that only certain areas or only a certain area of the thermoplastic textile connects to the surface. Other areas or another area do not connect, so that the permeability for air and/or humidity is maintained there, for example. The function and/or the design of the shoe upper of the soccer shoe according to the invention can be modified by this.

Flat Weft-Knitted Upper

[0207] FIG. 9 shows embodiments of an upper **91** according to the present invention. The upper **91** shown in FIG. 9 comprises flat weft-knitted knitwear **92**. The knitwear **92** can be manufactured on a suitable flat weft-knitting machine. In a flat weft-knitting machine, a thread feeder feeds the thread back and forth along one or several beds of needles.

[0208] The flat weft-knitted knitwear **92** of the upper **91** according to the invention forms a top portion **93** and a bottom portion **94** of the upper. The top portion **93** is positioned essentially above the bottom portion **94** when the sports shoe in which the upper **91** is used is standing. In other words, the top portion **93** is located closer to the edge of foot opening than the bottom portion **94**, with the foot opening being the opening through which a foot is inserted when the sports shoe is put on. The top portion **93** and the bottom portion **94** can be directly adjacent to one another or they can be spaced from one another. In the embodiments of FIG. 9, the top portion **93** and the bottom portion **94** are adjacent to one another.

[0209] The top portion **93** shown in FIG. 9 is formed as tube weft-knitted knitwear such that it is suitable for surrounding a part of the shank of the wearer of the sports shoe. Tube weft-knitted knitwear is two-ply knitwear which was manufactured on a flat weft-knitting machine with at least two beds of needles and the two plies of which are only connected on the edges. Thus, tube weft-knitted knitwear is created when the stitches created on a first bed of needles are weft-knitted to the stitches created on a second bed of needles merely on the edge of the weft-knitted fabric.

[0210] The bottom portion **94** of the upper **91** according to the invention is suitable for covering at least a part of a foot of the wearer of the sports shoe. In the embodiments of FIG. 9, the bottom portion **94** covers the foot essentially completely, i.e. from the toes to below the ankle, for example. However, in certain embodiments, the bottom portion **94** may not cover the entire foot or may at least partially not comprise knitwear but another material in another area. For example, the upper can comprise a front portion, e.g. in the area of the toes, which is not formed as knitwear but comprises leather or artificial leather, for example.

[0211] In the embodiments of FIG. 9, the knitwear **92** is formed as single-surface knitwear. Thus, the top portion **93** and the bottom portion **94** are weft-knitted on a flat weft-knitting machine in one go and connected to one another in the process in the area specified with reference number **95**.

However, in certain embodiments, the top portion **93** and the bottom portion **94** may be manufactured as separate knitwear and subsequently connected to one another by linking. In linking, two edges of knitwear are connected to each other according to the stitches (usually stitch by stitch). Due to this, no seam or at most a seam which only adds a little thickness is created at the connecting line **95** of the top portion **93** and the bottom portion **94**.

[0212] Alternatively, the top portion **93** and the bottom portion **94** can be manufactured separately and connected to one another by ultrasonic welding or other connecting techniques. Additionally, the welded seam can be covered by a band of adhesive material (e.g. thermoplastic or duroplastic (thermoset) material or adhesive, etc.). The band can be arranged on the inside of the upper **91**. Alternatively or additionally, the band can be arranged on the outside of the upper **91**.

[0213] In some embodiments, the top portion **93** and the bottom portion **94** may be sewn or glued together. The seam can also be covered by a band as described before in this case.

[0214] In the embodiments of FIG. **9**, the top portion **93** and the bottom portion **94** are adjacent to one another below an ankle (not shown in the Figure) of a wearer of the sports shoe in the area specified with reference number **95**. In certain embodiments, however, the top portion **93** and the bottom portion may be adjacent to one another in another area of the foot, e.g. above the ankle.

[0215] Since the embodiments of FIG. **9** are single-surface knitwear **92**, both the knitwear of the top portion **93** and that of the bottom portion **94** are weft-knitted on a flat weft-knitting machine with two beds of needles. In this regard, the top portion **93** is weft-knitted as a tube, i.e. the stitches of the two plies of the knitwear are only connected to one another at the edges. The bottom portion **94** is weft-knitted as a two-ply knitwear, whose plies are connected to one another, in the embodiments of FIG. **9**. When weft-knitting from the top end, i.e. from the opening for the foot, the two plies of the top portion **93**, which are only connected at the edge, therefore pass into two plies which are consistently connected to one another in the area of the transition **95** to the bottom portion **94**. If the upper is weft-knitted from its sole end, the two consistently connected plies of the bottom portion **94** pass into two plies of the top portion **93**, which are only connected at the edge, in the area of the transition **95** to the top portion **93**.

[0216] In the embodiments of FIG. **9**, the bottom portion is therefore formed as two-ply knitwear. If the top portion **93** and the bottom portion **94** are knitted separately and connected to one another subsequently, the knitwear of the bottom portion **94** can also be one-ply knitwear, i.e. knitwear which was weft-knitted on only one bed of needles.

[0217] In the embodiments of FIG. **9**, the top portion **93**, at its upper edge, comprises an elastic cuff **96** formed integrally with the knitwear **92**. The cuff **96** is therefore also knitwear. In this regard, the cuff **96** can e.g. comprise a particularly elastic knitting stitch (e.g. single Jersey) and additionally or alternatively comprise an elastic yarn, e.g. based on elastane. In certain embodiments, the cuff **96** may be manufactured separately and subsequently connected to the upper **91** by sewing, welding or linking.

[0218] In the embodiments of FIG. **9**, the top portion **93** is adjusted to the anatomical conditions of a shank of a wearer of the sports shoe. The human shank is tapered towards the ankle. Accordingly, the circumference of the top tube weft-knitted portion **93** increases from the transition area **95** to the portion **94** towards the top end, at which the cuff **96** is located. Thus, the top portion **93** is tapered from the top edge of the upper towards the ankle. Such a variation of the circumference can be realized on a flat weft-knitting machine by altering the number of stitches along the length of the top portion **93**.

[0219] In the embodiments of FIG. **9**, the top portion **93** is elastic and exerts pressure on at least a part of the shank. Elasticity can be caused by the type of knitting stitch. For example, the top portion **93** can be weft-knitted as single Jersey. Additionally or alternatively, an elastic yarn can also be used.

[0220] The exerted pressure can be adjusted to the sport for which the sports shoe is used. For example, the foot can, for example, be provided with better support by higher pressure in case of a tennis shoe, which is subject to high accelerations due to abrupt decelerations and accelerations. In case of a running shoe, which is usually not subject to such high accelerations as a tennis shoe is, less pressure on the shank is sufficient.

[0221] The exerted pressure can additionally or alternatively be adjusted to the respective wearer of the sports shoe. Depending on their personal preferences, the wearer can decide on more or less

pressure on the shank by the top portion **93**.

[0222] FIG. **10** shows embodiments of a sports shoe **101** according to the invention which comprises an upper **91** according to the invention and a sole **102** connected to the upper **91**. As shown in the embodiments of FIG. **10**, the sole **102** can comprise a midsole and an outsole. Additionally, the sports shoe **101** can comprise an insole (not shown in FIG. **10**). In certain embodiments, the sole may be a one-piece element fulfilling the function of an outsole, midsole and insole, if applicable. The sole **102** can e.g. be manufactured from EVA (ethylene vinyl acetate), rubber, extruded polyurethane (eTPU), extruded polyether block amide (ePEBA) or other plastics. In certain embodiments, the sole **102** may be manufactured from knitwear.

[0223] In the embodiments of FIG. **10**, the outsole is treaded. Depending on the use of the sports shoe **101**, the outsole can be strongly treaded, less strongly treaded or not treaded at all. If the sports shoe **101** is a soccer shoe, the sole **102** can comprise studs.

[0224] The sole **102** can be connected to the upper **91** e.g. by gluing, sewing or ultrasonic welding. If the sole **102** is manufactured from knitwear, the sole **102** can be weft-knitted integrally with the bottom portion **94** of the upper **91**.

[0225] In the embodiments of FIG. **10**, the bottom portion **94** comprises at least one lace bar **103**, formed integrally with the knitwear **92**, with at least one lace eyelet **104**. In certain embodiments, the lace bar **103** may be manufactured separately and connected to the upper **91** by e.g. sewing, gluing, welding or linking. In the embodiments of FIG. **10**, the lace bar is formed as a ply of the two-ply knitwear **92** of the bottom portion **94**. The second ply of the knitwear **92** assumes the function of a tongue.

[0226] In the embodiments of FIG. **10**, the at least one lace eyelet **104** is weft-knitted into the lace bar **103**. Thus, the lace eyelet **104** is formed as the lace bar **103** is weft-knitted. In certain embodiments, the lace eyelet **104** may be subsequently formed in the lace bar **103**, for example by punching. The lace bar **103** can comprise any desired number of lace eyelets.

[0227] The knitwear **92** of the upper **91** of FIG. **10** is also partially provided with a polymer coating. In the areas of which four are specified by reference number **105** by way of example, the knitwear **92** is not provided with a polymer coating. In the other areas, the knitwear **92** is provided with a polymer coating. The arrangement of the coated and uncoated areas can also differ from the arrangement shown in FIG. **10** on principle, or the knitwear **92** does not comprise any polymer coating.

[0228] The polymer coating can be a coating as described in the section “Polymer coating” and it can be processed as described therein. Instead of a polymer coating, a thermoplastic textile can also be used as reinforcement, as described in the corresponding section. The statements made with regard to the polymer coating then apply analogously with regard to the arrangement of reinforced and non-reinforced areas.

[0229] In the embodiments of FIG. **10**, the knitwear is coated with polymer especially along the back shank area from the top cuff **96** to the heel area, i.e. in the area of the Achilles' tendon, and along the area adjacent to the sole. The coated areas extend from there, finger-like, to the area of the shin or the instep. This arrangement of coated and uncoated areas allows the upper **91** to be provided with stability, on the one hand, and to maintain its flexibility when walking or running, on the other hand.

[0230] The upper **91** of FIG. **10** furthermore comprises a reinforcement **106** in the heel area. The reinforcement **106** can e.g. be manufactured from polyurethane, EVA or rubber. The reinforcement **106** can e.g. be glued or welded to the upper **91**. Additionally or alternatively, the upper **106** can comprise reinforcements in other areas, e.g. in the toe area.

[0231] FIG. **11** shows further embodiments of a sports shoe **101** according to the invention with an upper **91** according to the invention in a schematic representation. The sports shoe **101** shown in FIG. **11** is particularly suitable for sports like soccer or rugby, since its sole **102** comprises studs. On principle, however, the sports shoe **101** can also comprise no studs but only a tread or no tread.

[0232] In addition to the embodiments of FIG. 10, the upper **91** comprises a pocket **111** in FIG. 11. A shin guard can be inserted into the pocket **111**, for example. The pocket **111** can be manufactured from knitwear. The pocket **111** can be formed integrally with the knitwear **92** of the top portion **94**, i.e. it can be weft-knitted in one working step as a ply of the knitwear **92**. The pocket can, for example, be knitted in a third row of the weft-knitting machine, whereas the tube weft-knitted knitwear **92** of the top portion **94** is weft-knitted on the first and second rows of the knitwear. Alternatively, the pocket can also be weft-knitted as a third ply of the two-ply tube weft-knitted knitwear **92** of the top portion **93** with every other needle of a first or second row of a flat weft-knitting machine. Further alternatively, the pocket **111** is manufactured separately, e.g. as knitwear, and subsequently connected to the top portion, e.g. by sewing, gluing, welding or linking.

[0233] In the embodiments of FIG. 11, the top portion **93** comprises an inner clasp **112**. The clasp **112** e.g. enables a padding (e.g. on the basis of gel or foam) to be fastened, which pads the heel and increases comfort and stability. The clasp **112** can also be used for fastening a warming or cooling pack.

[0234] In the embodiments of FIG. 11, the top portion **93** and the bottom portion **94** are formed as single-surface knitwear **92**. Since in flat weft-knitting, transitions of more than 180° cannot be realized as connected knitwear, the first portion **93** and the second portion **94** are additionally connected in the heel area by a seam **113**. The seam **113** can be formed by linking of the top portion **93** and the bottom portion **94** or e.g. by ultrasonic welding. An advantage of linking is that the seam is not perceivable at all or hardly perceivable and that the knitting patterns of the portions are connected to one another stitch by stitch. In certain embodiments, the top portion **93** may be sewn to the bottom portion **94** in the area of the seam **103**.

[0235] FIG. 12 shows a schematic view of an upper **91** according to the invention. In these embodiments, the bottom portion **94** comprises two lace bars **103**. The lace bars **103** can be formed integrally with the knitwear **92** of the bottom portion **94**. In certain embodiments, however, the lace bars **103** may be manufactured separately and connected to the upper **91** by e.g. sewing, gluing, welding or linking.

[0236] In the embodiments of FIG. 12, the lace bars **103** are formed as a first ply of the two-ply knitwear **92** of the bottom portion **94**. The area **121** between the lace bars **103** is formed as two-ply knitwear **92** and assumes the function of a tongue. The area **121** can be more elastic than other areas of the upper. The area **121** can comprise an elastic stitch, e.g. single Jersey or double Jersey. Additionally or alternatively, an elastic yarn can also be used in the area **121**. Instead of single Jersey, rib fabric or purl fabric can also be used.

[0237] FIG. 12 shows the connection between the top portion **93** and the bottom portion **94**. As shown in FIG. 12, the top portion **93** is weft-knitted integrally with the bottom portion **94** in part. The integrally weft-knitted part **125** can constitute approximately 50% of the length of the connection between the top portion **93** and the bottom portion **94**. Alternatively, the integrally weft-knitted part **125** can constitute at least 25% of the length of the connection between the top portion **93** and the bottom portion **94**. As shown in FIG. 12, the unconnected part **123** of the top portion **93** can be connected to the bottom portion **94** using the edge parts **124**. The edges **122** can be connected by sewing, welding or any other known connecting method so as to form a seam near the heel. Alternatively, the edges **122** can be weft-knitted into shape as shown together or weft-knitted in a straight manner.

[0238] FIGS. 13a, 13b and 13c show further embodiments of an upper **91** according to the invention. In this regard, FIG. 13a shows a top view, FIG. 13b shows an inside view and FIG. 13c shows a side view.

[0239] In these embodiments, the upper **91** comprises an elastic instep area **131**. In the area of the forefoot, the knitwear comprises two areas **132** and **133**, with the stitch structure in the area **132** being different from the stitch structure of the area **133**. The lace bar **103** is formed integrally with the knitwear of the upper **91**. The top portion **93** of the upper **91** is formed as a tube weft-knitted

knitwear and comprises an elastic cuff **96**, which can be of different elasticity than the area **93** lying below it.

[0240] As shown in FIG. **13b**, the tube weft-knitted knitwear of the top portion **93** continues below the bottom portion **94** and forms a heel lining **134**, which at surrounds at least part of a heel of a wearer, within the upper **91**. Thus, the heel lining **134** provides additional padding. In the side view of FIG. **13c**, the course of the heel lining **134** in the area **135** is suggested by the dashed line. The upper **91** further comprises a heel pocket in the area specified with reference number **136**.

[0241] FIG. **14** shows further embodiments of an upper **91** according to the invention. As shown in that Figure, the upper **91** can comprise several zones with different properties. In the case of the upper **91** shown in FIG. **14**, the upper **91** can comprise a single layer when two beds of needles are used in the manufacture of the upper **91**. For example, weft-knitting the upper **91** can start at the cuff **96** and run along portions **141** and **142** to the bottom area **143** of the upper **91**. In this regard, area **141** comprises an elastic weft-knitted fabric, whereas area **142** comprises a rigid weft-knitted fabric.

[0242] The area **143** is an example of a strobel sole replacement which can be flat weft-knitted and then connected to the weft-knitted upper. For example, the area **143** can be connected with an upper by a seam near the dashed line. Furthermore, in some embodiments a weft-knitted insole and/or outsole can be provided and connected with the upper. These flat weft-knitted parts (e.g. strobel sole, insole and/or outsole) can include elastic yarns in some embodiments.

[0243] When manufacturing the upper **91**, every bed of needles can be used for weft-knitting a side of the upper **91**, for example the lateral side and the medial side. During knitting of the upper **91**, the sides of the upper **91** are connected with one another at the ends of the bed of needles. For example, the lateral side can be connected to the medial side by the stitches being transferred between the beds of needles, i.e. being shifted from one bed of needles to the respective other bed of needles. Once weft-knitting of the upper **91** is completed, the open stitches on the beds of needles can also be linked so as to form the upper **91**.

[0244] Alternatively, some embodiments of a weft-knitted upper **91** can be manufactured using additional beds of needles so as to increase the number of plies on one or both sides of the upper **91**. Additional layers can provide additional padding, allow the manufacture of structures (e.g. support, strips, bands, pockets, etc.) and allow the selection of predetermined properties in the shoe such as heat-transport properties, regulation of humidity, etc.

[0245] Instead of being manufactured by several beds of needles, additional layers can also be manufactured on a single bed of needles, by each nth (e.g. each second or third) needle being responsible for a single ply. Needles which are responsible for a certain ply can use a separate thread feeder and/or a different yarn. A different thread feeder could be used for the 1st, 3rd, 5th, etc. needle than for the 2nd, 4th, 6th, etc. needle, for example. A first ply of the knitwear is then manufactured on needles 1, 3, 5, etc. while a second ply of the knitwear is simultaneously manufactured on needles 2, 4, 6, etc.

[0246] In a further example, an upper can be weft-knitted using two or more beds of needles. A first bed of needles can be used for weft-knitting a first side of an upper **91** with a first length and the second bed of needles can be used for weft-knitting the second side of the upper **91** with a shorter second length. The two sides can also be connected with one another by linking. In some embodiments, weft-knitting can be continued after linking on at least one side, so as to obtain an additional weft-knitted portion which continues on from the linking area. This additional weft-knitted area can e.g. be used for providing additional support in a part of the upper, for enveloping the midsole and/or for creating structures (e.g. strips, bands, pockets) on the upper **91**.

[0247] FIGS. **15**, **16** and **17** show further embodiments of an upper according to the invention. The uppers **91** shown in FIGS. **15**, **16** and **17** comprise a tube weft-knitted top portion **93** and a bottom portion **94**. In the embodiments of FIGS. **16** and **17**, the upper **91** further comprises a portion **161**, which comprises flat weft-knitted, elastic knitwear. The portion **161** is partially arranged in the

instep area and assumes the function of a tongue there. The portion **161** overlaps with the bottom portion in the area **162**.

[0248] A method for manufacturing an upper according to the invention comprises the following steps: Flat weft-knitting the knitwear such that the flat weft-knitted knitwear forms a top portion and a bottom portion of the upper; forming the top portion as tube weft-knitted knitwear such that it is suitable for surrounding a part of the shank of a wearer of the sports shoe; and forming the bottom portion such that it is suitable for covering at least a part of a foot of the wearer of the sports shoe.

[0249] On principle, all described embodiments can be combined with one another, i.e. the features of one embodiment together with the features of another embodiment constitute a further embodiment of the invention.

[0250] In the following, further examples are described to facilitate the understanding of the invention: [0251] 1. Upper (**91**) for a sports shoe, comprising: [0252] flat weft-knitted knitwear (**92**), [0253] wherein the flat weft-knitted knitwear (**92**) forms a top portion (**93**) and a bottom portion (**94**) of the upper (**91**), [0254] wherein the top portion (**93**) is formed as tube weft-knitted knitwear such that it is suitable for surrounding a part of the shank of the wearer of the sports shoe, and [0255] wherein the bottom portion (**94**) is suitable for covering at least a part of a foot of the wearer of the sports shoe. [0256] 2. Upper (**91**) according to example 1, wherein the top portion (**93**) and the bottom portion (**94**) are connected to one another by linking and/or by weft-knitting. [0257] 3. Upper (**91**) according to example 1, wherein the top portion (**93**) and the bottom portion (**94**) are connected to one another by ultrasonic welding. [0258] 4. Upper (**91**) according to example 3, wherein the welded seam is covered by a band of adhesive material. [0259] 5. Upper (**91**) according to example 1, wherein the knitwear (**92**) is formed as single-surface knitwear. [0260] 6. Upper (**91**) according to any one of the preceding examples, wherein the top portion (**93**) and the bottom portion (**94**) are adjacent to one another below the ankle of the wearer of the sports shoe. [0261] 7. Upper (**91**) according to any one of the preceding examples, wherein the knitwear (**92**) is manufactured on a flat weft-knitting machine with two beds of needles. [0262] 8. Upper (**91**) according to any one of the preceding examples, wherein the bottom portion (**94**) is formed as two-ply knitwear. [0263] 9. Upper (**91**) according to any one of the preceding examples, wherein the top portion (**93**) comprises a weft-knitted-in pocket (**111**) for a shin guard. [0264] 10. Upper (**91**) according to any one of the preceding examples, wherein the bottom portion (**94**) comprises at least one lace bar (**103**), formed integrally with the knitwear (**92**), with at least one lace eyelet (**104**). [0265] 11 Upper (**91**) according to example 10, wherein the lace bar (**103**) is formed as a layer of the knitwear (**92**). [0266] 12. Upper (**91**) according to any one of examples 10 or 11, wherein the bottom portion (**94**) comprises two lace bars (**103**) and the knitwear (**92**) is more elastic in an area (**121**) between the two lace bars (**103**) than in other areas. [0267] 13. Upper (**91**) according to any one of the preceding examples, further comprising a front portion which is not formed as knitwear. [0268] 14. Upper (**91**) according to example 13, wherein the front portion comprises leather or artificial leather. [0269] 15. Upper (**91**) according to any one of the preceding examples, wherein the top portion (**93**), at its upper edge, comprises an elastic cuff (**96**) formed integrally with the knitwear. [0270] 16. Upper (**91**) according to any one of the preceding examples, wherein the top portion (**93**) is adjusted to the anatomical conditions of a shank of a wearer of the sports shoe. [0271] 17. Upper (**91**) according to any one of the preceding examples, wherein the top portion (**93**) is tapered from an upper edge to an ankle area. [0272] 18. Upper (**91**) according to any one of the preceding examples, wherein the top portion (**93**) is elastic and exerts pressure on at least a part of the shank. [0273] 19. Upper (**91**) according to example 18, wherein the exerted pressure is adjusted to the sport for which the sports shoe is used. [0274] 20. Upper (**91**) according to one of examples 18 or 19, wherein the exerted pressure is adjusted to the respective wearer of the sports shoe. [0275] 21. Upper (**91**) according to one of examples 18 through 20, wherein the top portion (**93**) comprises an elastic yarn. [0276] 22. Sports shoe (**101**) comprising:

[0277] an upper (91) according to one of examples 1 through 21; and [0278] a sole (102) which is connected to the upper (91). [0279] 23. A method of manufacturing a shoe upper (91) according to one of examples 1 through 21, comprising the following steps: [0280] flat-knitting the knitwear (92) such that the flat weft-knitted knitwear (92) forms a top portion (93) and a bottom portion (94) of the upper (91); [0281] forming the top portion (93) as tube weft-knitted knitwear (92) such that it is suitable for surrounding a part of the shank of the wearer of the sports shoe; and [0282] forming the bottom portion (94) such that it is suitable for covering at least a part of a foot of the wearer of the sports shoe.

[0283] Different arrangements of the components depicted in the drawings or described above, as well as components and steps not shown or described are possible. Similarly, some features and sub-combinations are useful and may be employed without reference to other features and sub-combinations. Embodiments of the invention have been described for illustrative and not restrictive purposes, and alternative embodiments will become apparent to readers of this patent. Accordingly, the present invention is not limited to the embodiments described above or depicted in the drawings, and various embodiments and modifications may be made without departing from the scope of the claims below.

Claims

1-20. (canceled)

21. An upper for a sports shoe comprising: a knitted element comprising: a top portion comprising multi-ply knitwear; a bottom portion comprising multi-ply knitwear, wherein at least two plies are consistently connected to one another and cover at least a portion of at least one of a top, sides, or a heel of a foot of the wearer of the sports shoe when the sports shoe is worn; and wherein the knitted element comprises at least one of a weft or a filler yarn in the multi-ply knitwear of at least one of the top or bottom portion.

22. The upper for a sports shoe of claim 21, wherein the top portion comprises at least two plies are separated from each other and joined at edges to form a tube that surrounds at least part of an ankle of a wearer of the sports shoe when the sport shoe is worn.

23. The upper for a sports shoe of claim 22, wherein the at least one of a weft or a filler yarn in the multi-ply knitwear of at least one of the top or bottom portion reduces stretchability of at least one of the top or bottom portion.

24. The upper of claim 22, wherein the bottom portion substantially covers a top and sides of a foot of the wearer of the sport shoe when the sports shoe is worn.

25. The upper of claim 22, wherein the top portion and the bottom portion are adjacent to one another below the ankle of the wearer of the sports shoe when worn.

26. The upper of claim 22 wherein the bottom portion and the top portion are unitarily knitted together in a single knitting process.

27. The upper of claim 22 wherein the knitted element comprises a single layer.

28. The upper of claim 27 wherein the knitted element is single layer weft-knitted.

29. The upper of claim 27 or 28 wherein the knitted element is flat knitted.

30. The upper of claim 22, wherein the knitted element is formed as a single-surface knitwear.

31. The upper of claim 22, wherein the knitted element comprises a monofilament yarn.

32. The upper of claim 22, further comprising a front portion, wherein the knitted element excludes the front portion, and wherein the front portion comprises leather or artificial leather.

33. The upper of claim 22, wherein the top portion, at its upper edge, comprises an elastic cuff formed integrally with the knitted element.

34. The upper of claim 22, wherein the top portion is adjusted to anatomical conditions of the at least part of the ankle of the wearer of the sports shoe.

35. The upper of claim 22, wherein the top portion is elastic, and wherein the elastic is configured

to exert pressure on the at least part of the ankle.

36. The upper of claim 35, wherein a configuration of the elastic results in the pressure being specific to a sport for which the sports shoe is used or being respective to the wearer.

37. A method of manufacturing a shoe upper for a sports shoe, comprising: flat knitting a knitted element, wherein knitting the knitted element comprises: forming a top portion comprising multiply knitwear; and performing at least one of placing a length of a thread between two wales in longitudinal direction to define a filler yarn or introducing a weft into the knitwear.

38. The method of claim 37, wherein forming the top portion comprises forming a top portion in which at least two plies are separated from each other and joined at edges to form a tube that surrounds at least part of an ankle of a wearer of the sports shoe when the sport shoe is worn.

39. The method of claim 38 wherein the method comprises placing the length of the thread between two wales in the longitudinal direction and holding the thread by transverse threads or other weave elements of the knitted element.

40. The method of claim 38, wherein only one bed of needles is used when flat knitting a knitted element, creating a one layered weft-knitted fabric.

41. The method of claim 38, wherein a monofilament yarn is used when flat knitting the knitted element.
